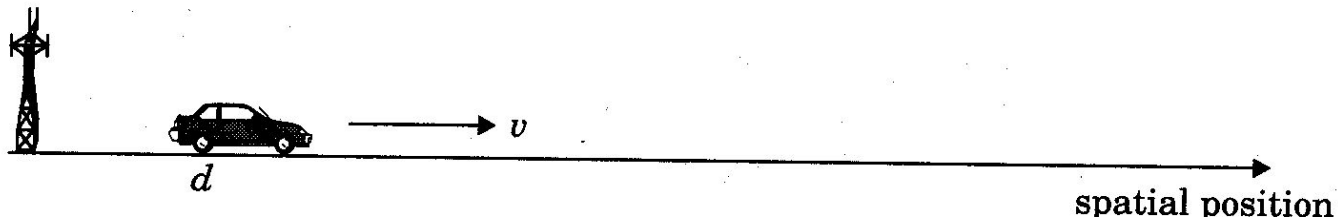


Mobile Radio Propagation: Small-Scale Fading and Multipath

Impulse Response Model of a Multipath Channel

- A mobile radio channel may be modeled as a linear filter with a time varying impulse response
 - time variation is due to receiver motion in space
 - Mobile radio channel can be modeled as Linear filter with a time varying impulse response.
 - Impulse response is a wideband channel characterization.



- The channel impulse response can be expressed as $h(d, t)$. Let $x(t)$ represent the transmitted signal, then the received signal $y(d, t)$ at position d can be expressed as

$$y(d, t) = x(t) \otimes h(d, t) = \int_{-\infty}^{\infty} x(\tau) h(d, t - \tau) d\tau$$



- For a causal system $h(d, t) = 0$ for $t < 0$

$$y(d, t) = \int_{-\infty}^t x(\tau) h(d, t - \tau) d\tau$$

- The position of the receiver can be expressed as

$$d = vt$$

- We have

$$y(vt, t) = \int_{-\infty}^t x(\tau) h(vt, t - \tau) d\tau$$

- Since v is a constant, $y(vt, t)$ is just a function of t .

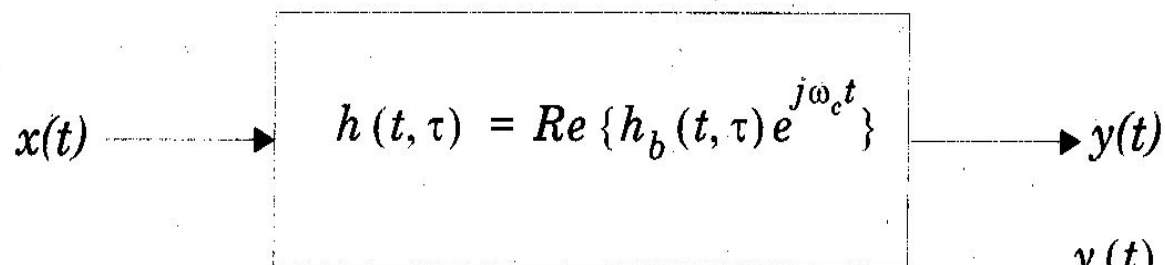
$$y(t) = \int_{-\infty}^t x(\tau) h(vt, t - \tau) d\tau$$

- In general, the channel impulse response can be expressed $h(t, \tau)$
 - t : time variation due to motion
 - τ : channel multipath delay for a fixed value of t .
- With the channel impulse response $h(t, \tau)$, we may have the output

$$y(t) = \int_{-\infty}^t x(\tau) h(t, \tau) d\tau = x(t) \otimes h(t, \tau)$$

- For bandlimited bandpass channel, then $h(t, \tau)$ may be equivalently described by a complex baseband impulse response $h_b(t, \tau)$ ★
 - The equivalent baseband output

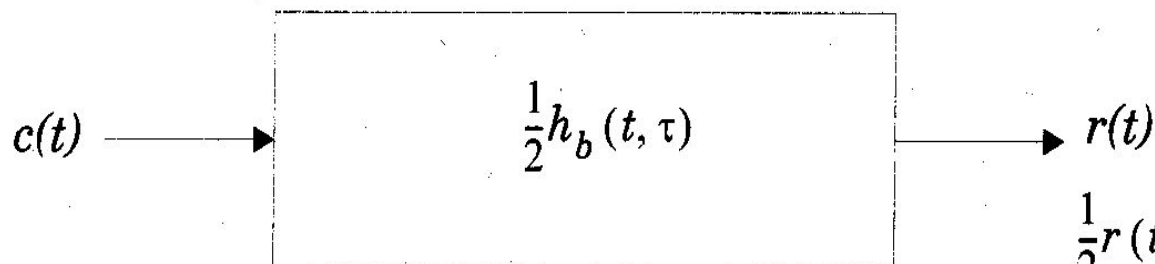
$$\frac{1}{2}r(t) = \frac{1}{2}c(t) \otimes \frac{1}{2}h_b(t, \tau) \quad \text{or} \quad r(t) = \frac{1}{2}c(t) \otimes h_b(t, \tau)$$



(a)

$$y(t) = \text{Re}\{r(t)e^{j\omega_c t}\}$$

$$y(t) = x(t) \otimes h(t)$$



(b)

$$\frac{1}{2}r(t) = \frac{1}{2}c(t) \otimes \frac{1}{2}h_b(t)$$

$$x(t) = \text{Re}\{c(t)\exp(j\omega_c t)\}$$

$$y(t) = \text{Re}\{r(t)\exp(j\omega_c t)\}$$

$$r(t) = \frac{1}{2}c(t) \otimes h_b(t, \tau)$$

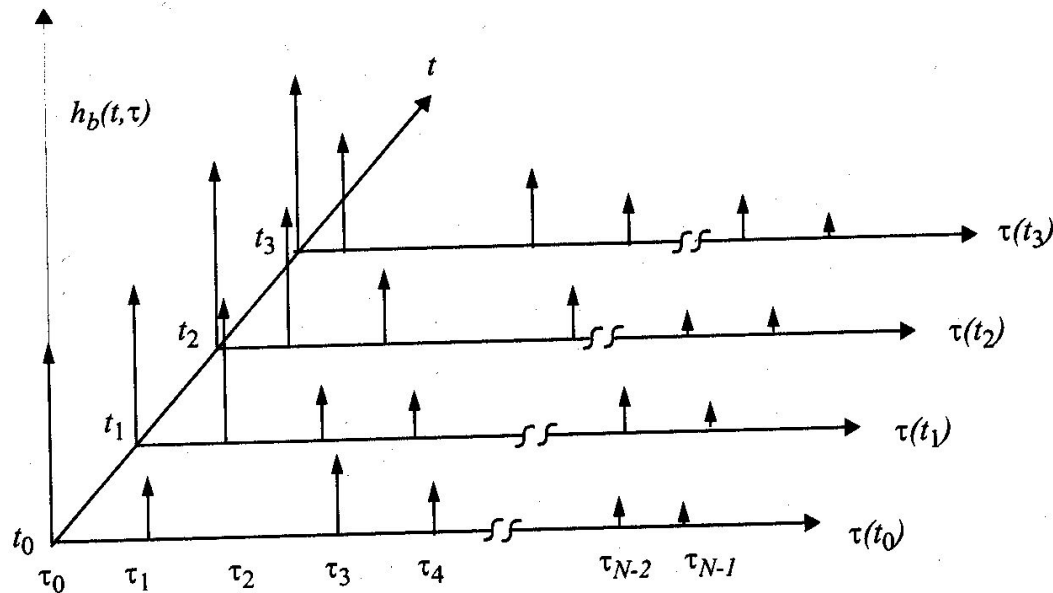
- Discretize the multipath delay axis τ into equal time delay segments called *excess delay bins*.
- The baseband response of a multipath channel can be expressed as

$$h_b(t, \tau) = \sum_{i=0}^{N-1} a_i(t, \tau) \exp(j2\pi f_c \tau_i(t) + j\phi(t, \tau)) \delta(\tau - \tau_i(t))$$

$a_i(t, \tau)$: amplitude of the i th multipath component

$\tau_i(t)$: excess delay of i th multipath component

- Define $\theta_i(t, \tau) = 2\pi f_c \tau_i(t) + \phi(t, \tau)$



- If the channel impulse response is assumed to be time invariant, the channel impulse response may be simplified as

$$h_b(\tau) = \sum_{i=0}^{N-1} a_i \exp(j\theta_i) \delta(\tau - \tau_i)$$

- The impulse response may be measured by using a probing pulse $p(t)$ which approximates a delta function.

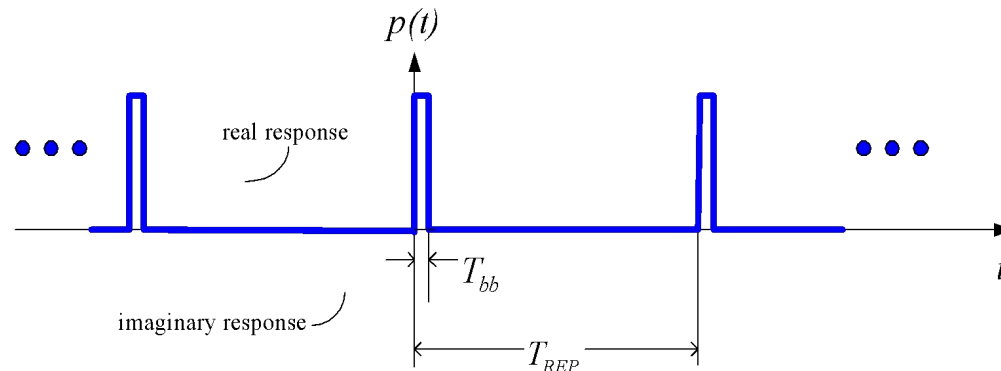
$$p(t) \approx \delta(t - \tau)$$

- Discrete time impulse response model for multipath channel can be used to analyse RF transmitted signal with **BW < (2/ΔΓ)**

4.2.1 Relationship Between Bandwidth and Received Power

- Consider a pulsed, transmitted signal of the form

$$x(t) = \text{Re}\{p(t) \exp(j2\pi f_c t)\}$$



- The signal $p(t)$ is a repetitive baseband pulse train with very narrow pulse width T_{bb} and repetition period T_{REP} , with $T_{REP} \gg \tau_{\max}$.
- Now, let

$$p(t) = 2\sqrt{\tau_{\max} / T_{bb}} \quad 0 \leq t \leq T_{bb}$$

- The channel output $r(t)$ closely approximates the impulse response and is given by

$$\begin{aligned}
 r(t) &= \frac{1}{2} \sum_{i=0}^{N-1} a_i \exp(-j\theta_i) \cdot p(t - \tau_i) \\
 &= \sum_{i=0}^{N-1} a_i \exp(-j\theta_i) \cdot \sqrt{\frac{\tau_{\max}}{T_{bb}}} \text{rect}\left[t - \frac{T_{bb}}{2} - \tau_i\right]
 \end{aligned}$$

- Instantaneous multipath power delay profile

$$\begin{aligned}
 |r(t_0)|^2 &= \frac{1}{\tau_{\max}} \int_0^{\tau_{\max}} r(t) r^*(t) dt \\
 &= \frac{1}{\tau_{\max}} \int_0^{\tau_{\max}} \frac{1}{4} \text{Re} \left\{ \sum_{j=0}^{N-1} \sum_{i=0}^{N-1} a_j(t_0) a_i(t_0) p(t - \tau_j) p(t - \tau_i) \exp(-j(\theta_j - \theta_i)) \right\} dt
 \end{aligned}$$

- If all the multipath components are resolved by the probe $p(t)$, then

$$|\tau_j - \tau_i| > T_{bb} \quad \forall \quad j \neq i$$

- Then we have

$$\begin{aligned} |r(t_0)|^2 &= \frac{1}{\tau_{\max}} \int_0^{\tau_{\max}} \frac{1}{4} \left(\sum_{k=0}^{N-1} a_k^2(t_0) p^2(t - \tau_k) \right) dt \\ &= \frac{1}{\tau_{\max}} \sum_{k=0}^{N-1} a_k^2(t_0) \int_0^{\tau_{\max}} \left(\sqrt{\frac{\tau_{\max}}{T_{bb}}} \text{rect} \left[t - \frac{T_{bb}}{2} - \tau_k \right] \right) dt \\ &= \sum_{k=0}^{N-1} a_k^2(t_0) \end{aligned}$$

- The total receiving power is related to the sum of the powers in the individual multipath components.

- Assuming that the received power from the multipath components forms a random process where each component has a random amplitude and phase at any time t , the average small-scale received power is

$$E_{a,\theta}[P_{WB}] = E_{a,\theta}\left[\sum_{i=0}^{N-1} |a_i \exp(j\theta_i)|^2\right] = \sum_{i=0}^{N-1} \overline{a_i^2}$$

- Now, consider a CW signal which is transmitted into the exact same channel, and let the complex envelope be given by $c(t)=2$. Then the received signal can be expressed as

$$r(t) = \sum_{i=0}^{N-1} a_i \exp(j\theta_i(t, \tau))$$

- The instantaneous power is ^{$i=0$} given by

$$|r(t)|^2 = \left| \sum_{i=0}^{N-1} a_i \exp(j\theta_i(t, \tau)) \right|^2$$

- In a local area, a_i varies little, but θ_i will vary greatly due to changes in propagation distance over space, resulting in large fluctuations of $r(t)$.
- The average received power over a local area is given by

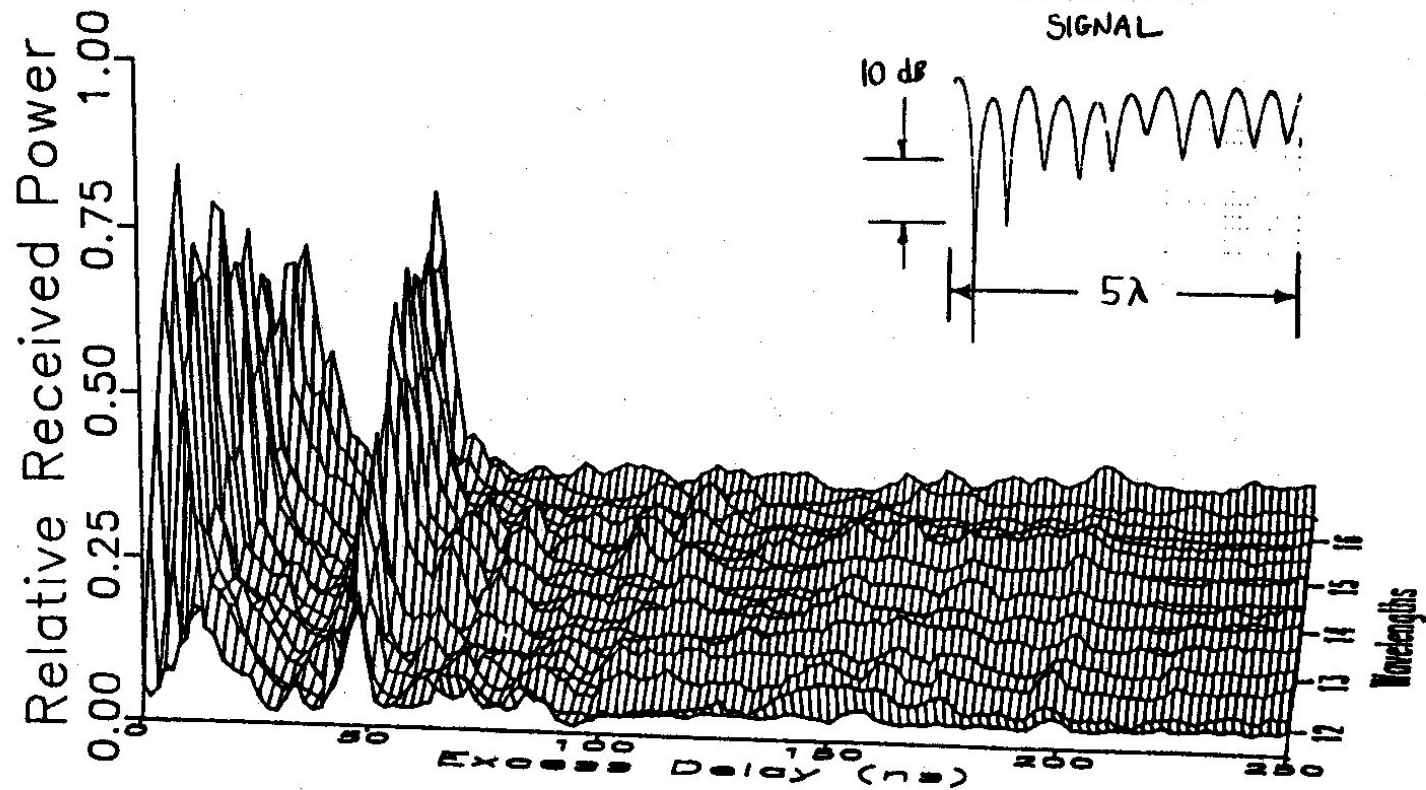
$$E_{a,\theta}[P_{CW}] = E_{a,\theta} \left[\left| \sum_{i=0}^{N-1} a_i \exp(j\theta_i(t, \tau)) \right|^2 \right]$$

$$\approx \sum_{i=0}^{N-1} \overline{a_i^2} + 2 \sum_{i=0}^{N-1} \sum_{i,j \neq i}^N r_{ij} \overline{\cos(\theta_i - \theta_j)}$$

where

$$r_{ij} = E_a[a_i a_j]$$

- The received power for CW wave has large fluctuations than that for WB signal.

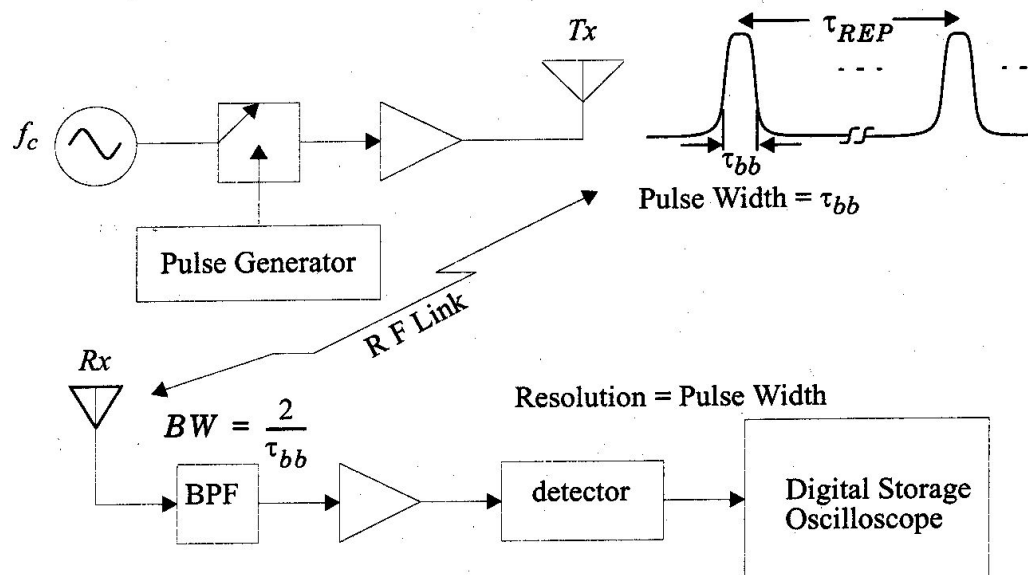


4.3 Small-Scale Multipath Measurement

- Multipath channel measurement techniques
 - Direct pulse measurements
 - Spread spectrum sliding correlator measurements
 - Swept frequency measurements

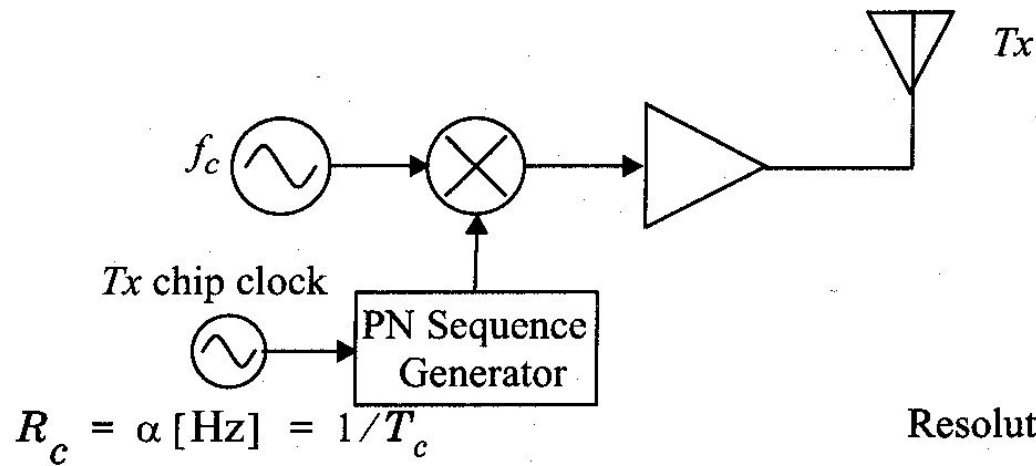
4.3.1 Direct RF Pulse System

- Direct RF pulse system
 - This system transmits a repetitive pulse of width τ_{bb} , and uses a receiver with a wideband filter with bandwidth $BW = 2 / \tau_{bb}$
 - Envelope detector to detect the amplitude response.
- Minimum resolvable delay τ_{bb}
- No phase information can be measured.

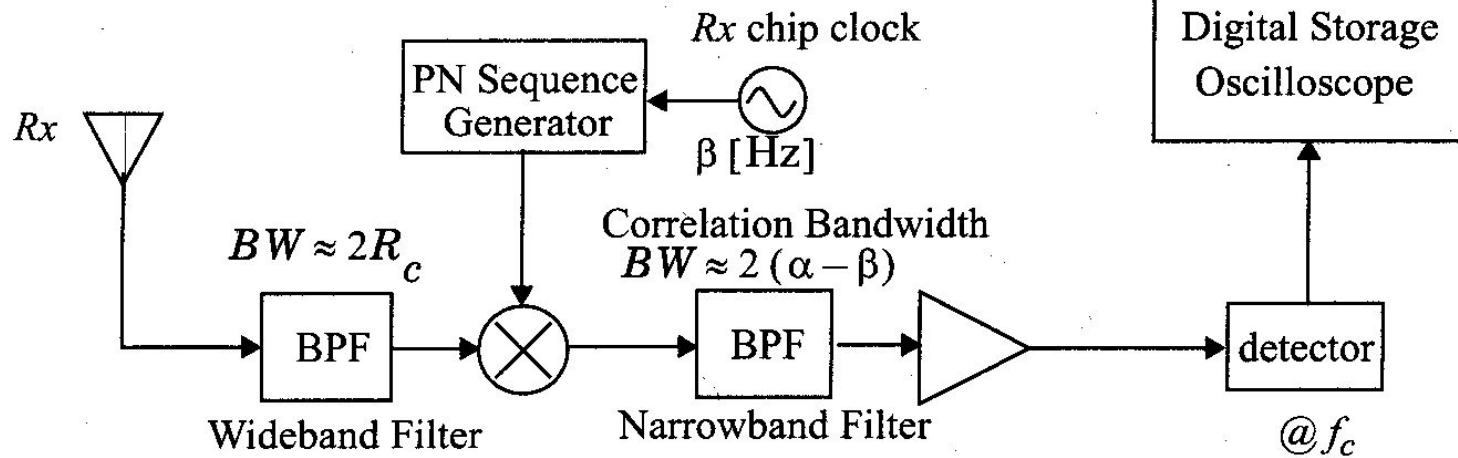


4.3.2 Spread Spectrum Sliding Correlator Channel Sounding

- System description
 - A carrier is spread over a large bandwidth by using a pseudo-noise sequence having chip duration T_c and a chip rate R_c .
 - Despread using a PN sequence identical to that used at the transmitter.
- The probing signal is wideband.
- Use a narrowband receiver preceded by a wideband mixer.
- The transmitter chip clock is run at a slightly faster rate than the receiver chip clock – sliding correlator.



$$\text{Resolution} \sim \frac{1}{R_c} \text{ (rms pulse width)}$$



- The time resolution of multipath components using a spread spectrum system with sliding correlation is

$$\Delta\tau = 2T_c = \frac{1}{R_c}$$

- The time between maximum correlation can be calculated

$$\Delta T = T_c r l = \frac{r l}{R_c}$$

T_c : chip period

r : sliding factor

R_c : chip rate

l : sequence length

- The sliding factor can be expressed as

$$r = \frac{\alpha}{\alpha - \beta}$$

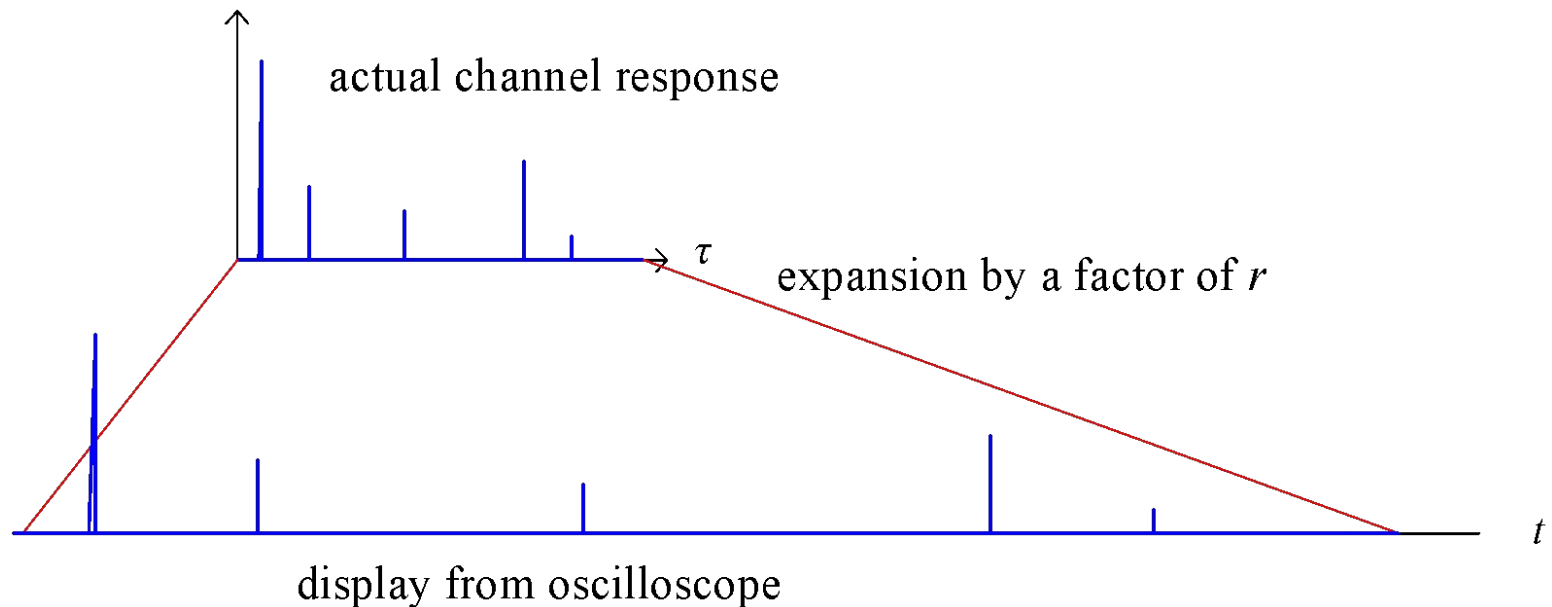
α : transmitter chip clock rate

β : receiver chip clock rate

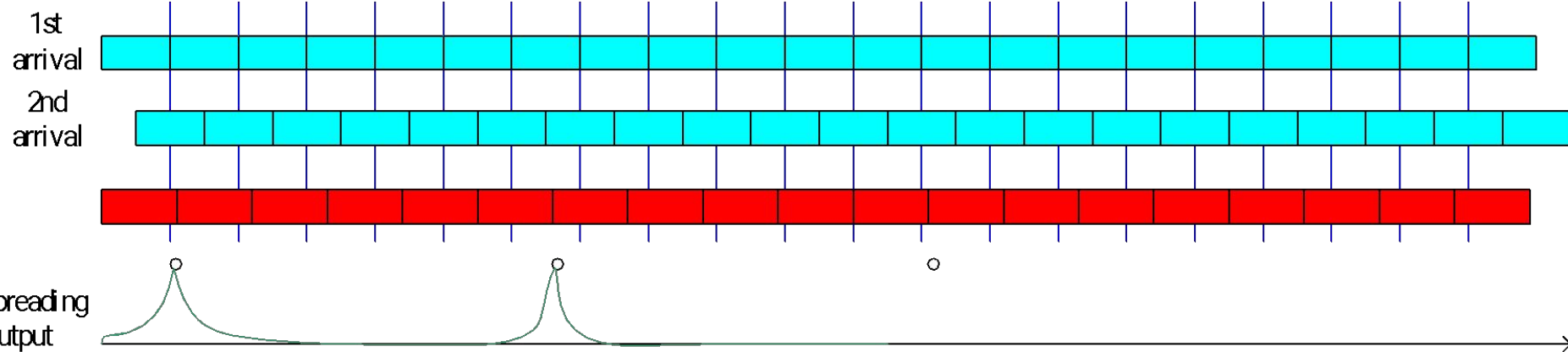
- The incoming signal is mixed with a PN sequence that is slower than the transmitter sequence. The signal is down converted to a low-frequency narrow band signal.

- The observed time scale on the oscilloscope using a sliding correlator is related to the actual propagation time scale by

$$\text{Actual Propagation Time} = \frac{\text{Observed Time}}{r}$$

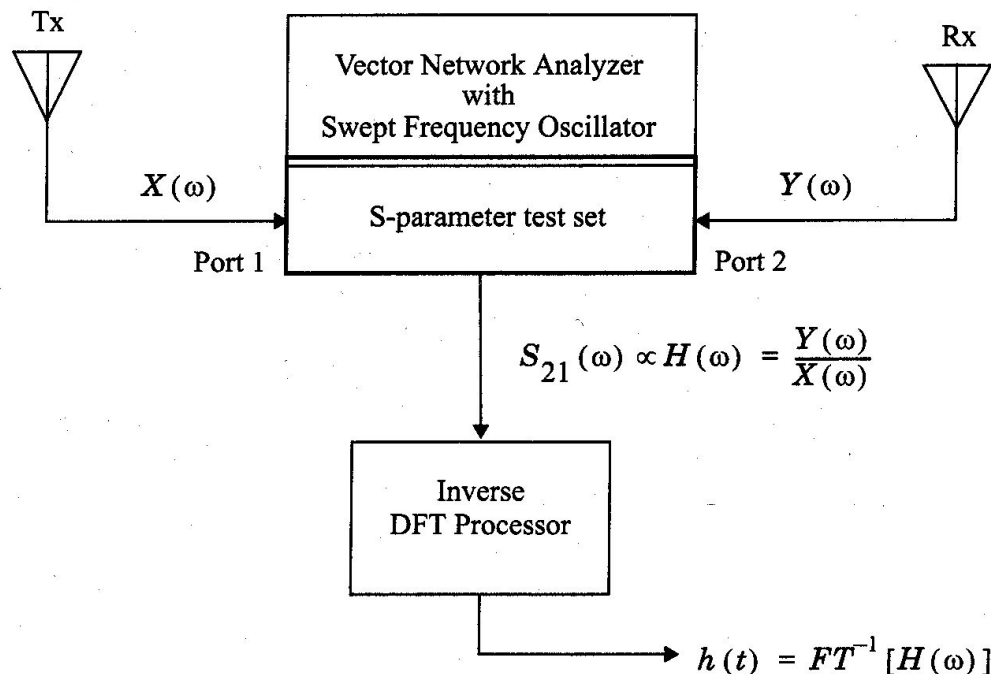


↑↑ channel response



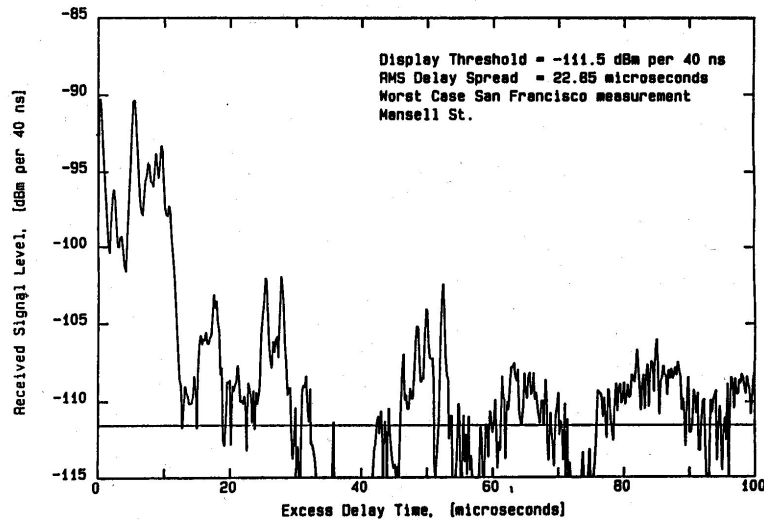
4.3.3 Frequency Domain Channel Sounding

- Dual relationship between time domain and frequency domain.
- It is possible to measure the channel impulse response in the frequency domain.
- Measure the frequency domain response and then converted to the time domain using inverse discrete Fourier transform (IDFT).

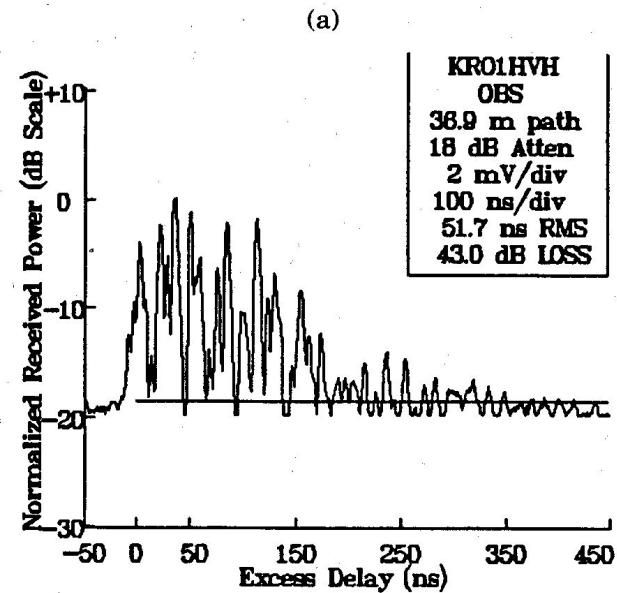


4.4 Parameters of Mobile Multipath Channels

- Power delay profiles for different types of channels are different



Outdoor



Indoor

4.4.1 Time Dispersion Parameters

- Time dispersion parameters
 - mean excess delay
 - RMS delay spread
 - excess delay spread
- Mean excess delay

$$\bar{\tau} = \frac{\sum_k a_k^2 \tau_k}{\sum_k a_k^2} = \frac{\sum_k P(\tau_k) \tau_k}{\sum_k P(\tau_k)}$$

- RMS delay spread

$$\sigma_{\tau} = \sqrt{\overline{\tau^2} - (\bar{\tau})^2}$$

where

$$\overline{\tau^2} = \frac{\sum_k a_k^2 \tau_k^2}{\sum_k a_k^2} = \frac{\sum_k P(\tau_k) \tau_k^2}{\sum_k P(\tau_k)}$$

- Depends only on the relative amplitude of the multipath components.
- Typical RMS delay spreads
 - Outdoor: on the order of microseconds
 - Indoor: on the order of nanoseconds
- Maximum excess delay (X dB) is defined to be the time delay during which multipath energy falls to X dB below the maximum.

$$\text{excess delay} = \tau_X - \tau_0$$

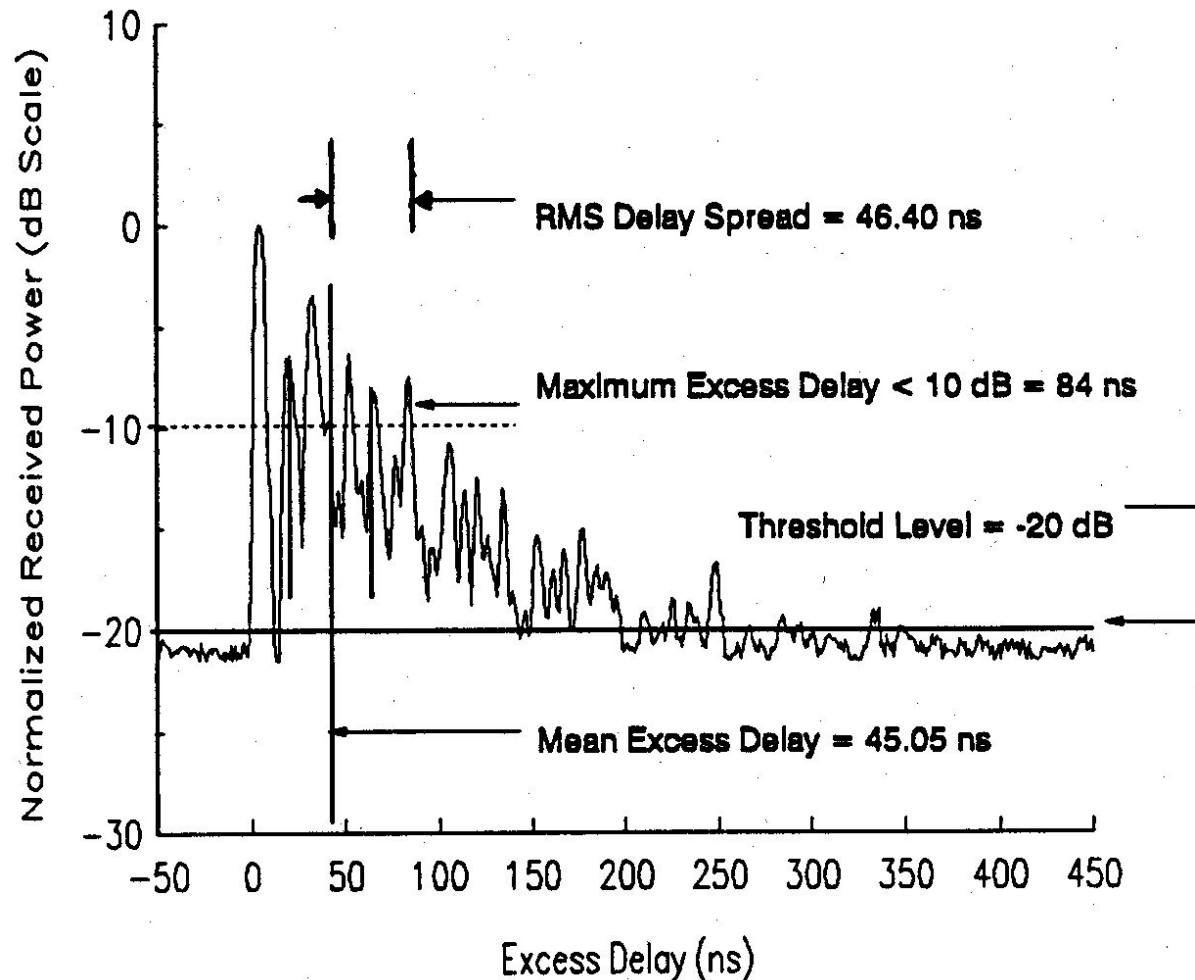
τ_X : maximum delay at which a multipath component is within X dB

τ_0 : delay for the first arriving signal

Table 4.1 Typical Measured Values of RMS Delay Spread

Environment	Frequency (MHz)	RMS Delay Spread (σ_τ)	Notes	Reference
Urban	910	1300 ns avg. 600 ns st. dev. 3500 ns max.	New York City	[Cox75]
Urban	892	10-25 μ s	Worst case San Francisco	[Rap90]
Suburban	910	200-310 ns	Averaged typical case	[Cox72]
Suburban	910	1960-2110 ns	Averaged extreme case	[Cox72]
Indoor	1500	10-50 ns 25 ns median	Office building	[Sal87]
Indoor	850	270 ns max.	Office building	[Dev90a]
Indoor	1900	70-94 ns avg. 1470 ns max.	Three San Francisco buildings	[Sei92a]

- Example of an indoor power delay profile; rms delay spread, mean excess delay, maximum excess delay (10dB), and the threshold level are shown



4.4.2 Coherent Bandwidth

- Coherent bandwidth, B_c , is a statistic measure of the range of frequencies over which the channel can be considered to be “flat”.
- Two sinusoids with frequency separation greater than B_c are affected quite differently by the channel.
- If the coherent bandwidth is defined as the bandwidth over which the frequency correlation function is above 0.9, then the coherent bandwidth is approximately

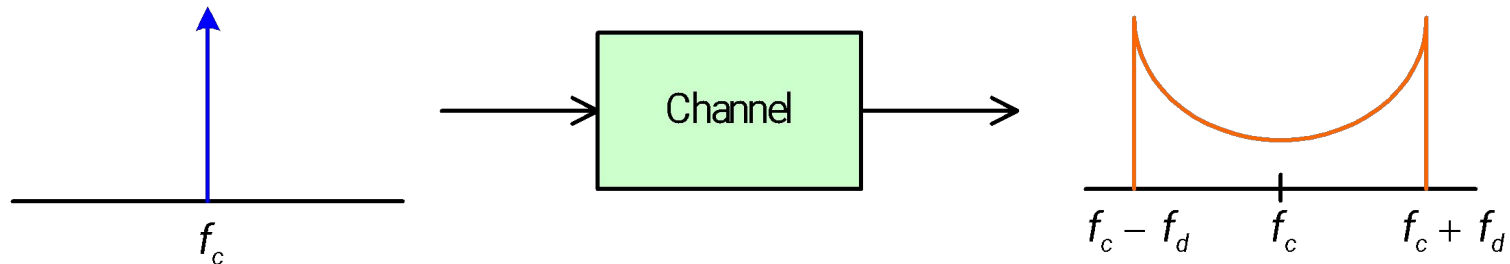
$$B_c \approx \frac{1}{50\sigma_\tau}$$

- If the frequency correlation function is above 0.5

$$B_c \approx \frac{1}{5\sigma_\tau}$$

4.4.3 Doppler Spread and Coherent Time

- Doppler spread and coherent time are parameters which describe the time varying nature of the channel in a small-scale region.
- When a pure sinusoidal tone of f_c is transmitted, the received signal spectrum, called the Doppler spectrum, will have components in the range $f_c - f_d$ and $f_c + f_d$, where f_d is the Doppler shift.



- f_d is a function of the relative velocity of the mobile, and the angle between the direction of motion of the mobile and direction of arrival of the scattered waves

- Coherent time T_c is the time domain dual of Doppler spread.
- Coherent time is used to characterize the time varying nature of the frequency dispersiveness of the channel in the time domain.

$$T_c \approx \frac{1}{f_m}$$

f_m : maximum Doppler shift given by $f_m = v / \lambda$

v : speed of the mobile λ : speed of the light

- Two signals arriving with a time separation greater than T_c are affected differently by the channel
- A statistic measure of the time duration over which the channel impulse response is essentially invariant.
- If the coherent time is defined as the time over which the time correlation function is above 0.5, then

$$T_c \approx \frac{9}{16\pi f_m}$$

4.4 Types of Small-Scale Fading

- Multipath delay spread leads to *time dispersion* and *frequency selective fading*.
- Doppler spread leads to *frequency dispersion* and *time selective fading*.
- Multipath delay spread and Doppler spread are independent of one another.

Small-Scale Fading

(Based on multipath time delay spread)

Flat Fading

1. BW of signal < BW of channel
2. Delay spread < Symbol period

Frequency Selective Fading

1. BW of signal > BW of channel
2. Delay spread > Symbol period

Small-Scale Fading

(Based on Doppler spread)

Fast Fading

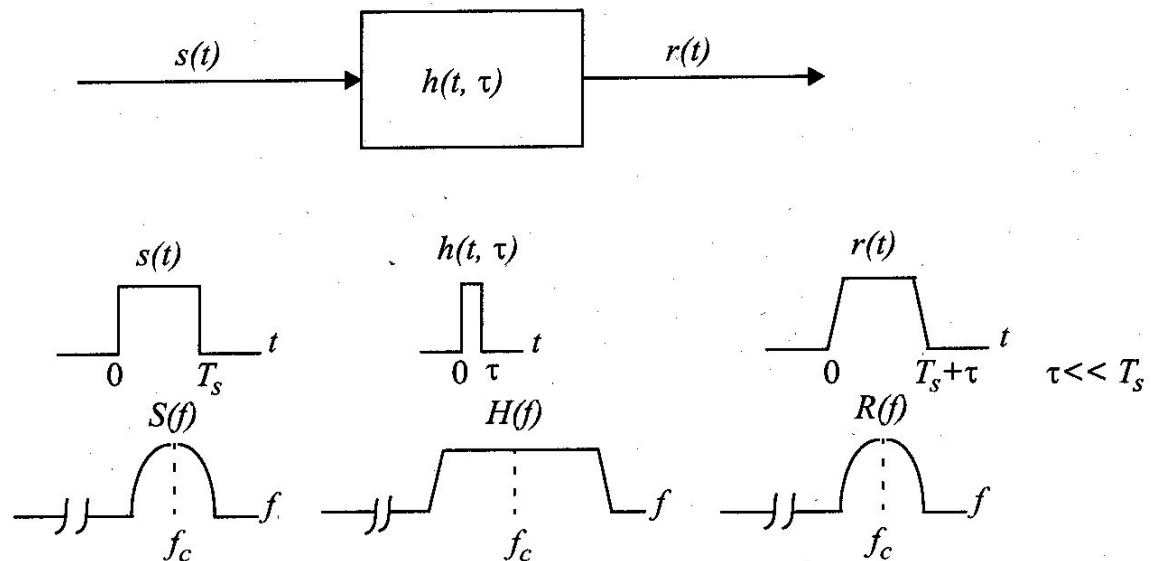
1. High Doppler spread
2. Coherence time < Symbol period
3. Channel variations faster than baseband signal variations

Slow Fading

1. Low Doppler spread
2. Coherence time > Symbol period
3. Channel variations slower than baseband signal variations

4.5.1 Flat Fading

- If the channel has a constant gain and linear phase response over a bandwidth which is greater than the bandwidth of the transmitted signal, the received signal will undergo **flat fading**.
- The received signal strength changes with time due to fluctuations in the gain of the channel caused by multipath.
- The received signal varies in gain but the spectrum of the transmission is preserved.



- Flat fading channel is also called **amplitude varying channel**.
- Also called **narrow band channel**: bandwidth of the applied signal is *narrow* as compared to the channel bandwidth.
- Time varying statistics: Rayleigh flat fading.
- A signal undergoes flat fading if

$$B_s \ll B_c$$

and

$$T_s \gg \sigma_\tau$$

T_s : reciprocal bandwidth (symbol period)

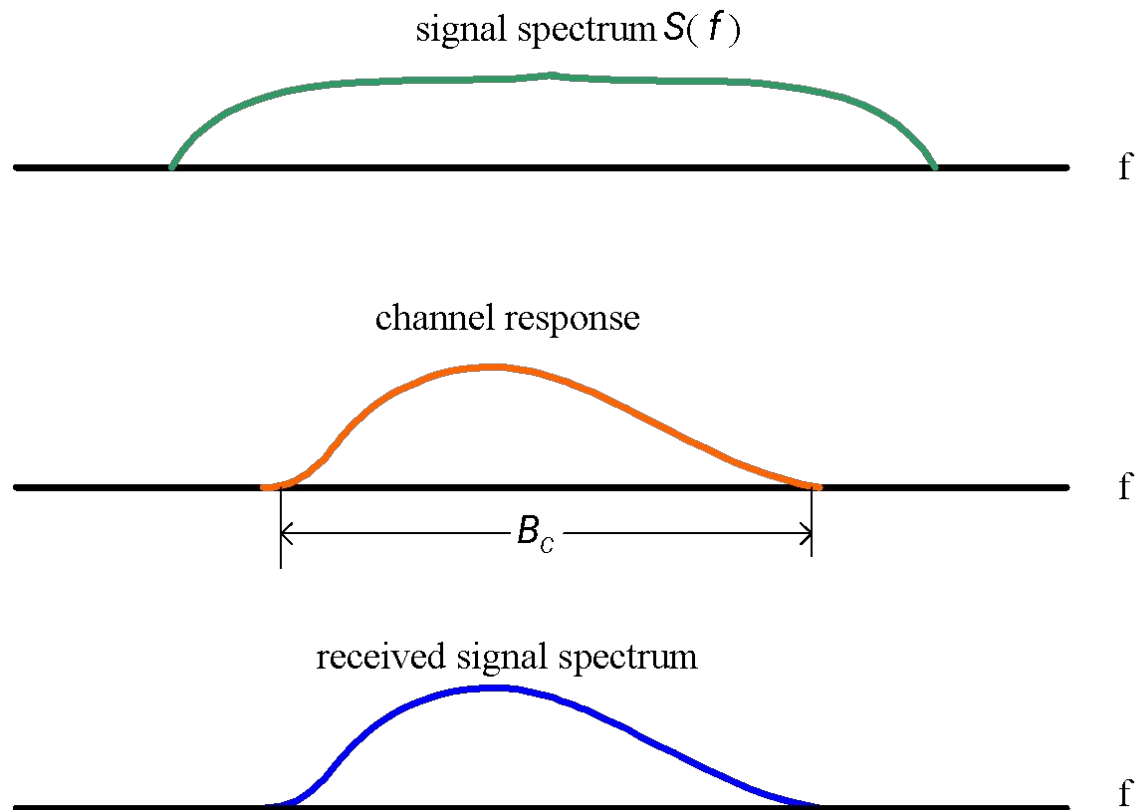
B_s : bandwidth of the transmitted signal

B_c : coherent bandwidth

σ_τ : rms delay spread

4.5.1 Frequency Selective Fading

- If the channel possesses a constant-gain and linear phase response over a bandwidth that is smaller than the bandwidth of transmitted signal, then the channel creates frequency selective fading.



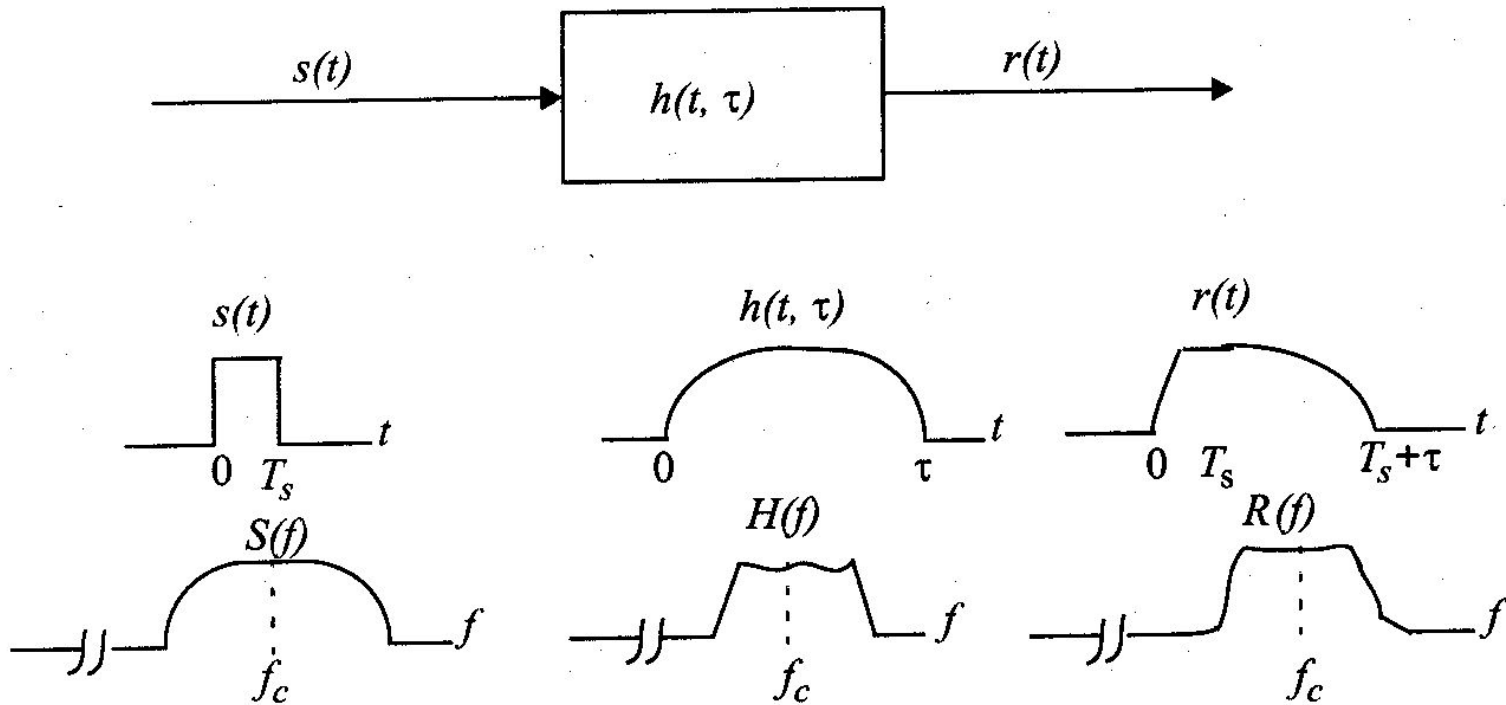
- Frequency selective fading is due to time dispersion of the transmitted symbols within the channel.
 - Induces intersymbol interference
- Frequency selective fading channels are much more difficult to model than flat fading channels.
- Statistic impulse response model
 - 2-ray Rayleigh fading model
 - computer generated
 - measured impulse response
- For frequency selective fading

and

$$B_s > B_c$$

$$T_s > \sigma_\tau$$

- Frequency selective fading channel characteristic



4.5.2 Fading Effects Due to Doppler Spread

- Fast Fading: The channel impulse response changes rapidly within the symbol duration.
 - The coherent time of the channel is smaller than the symbol period of the transmitted signal.
 - Cause frequency dispersion due to Doppler spreading.
- A signal undergoes fast fading if

$$T_s > T_c$$

and

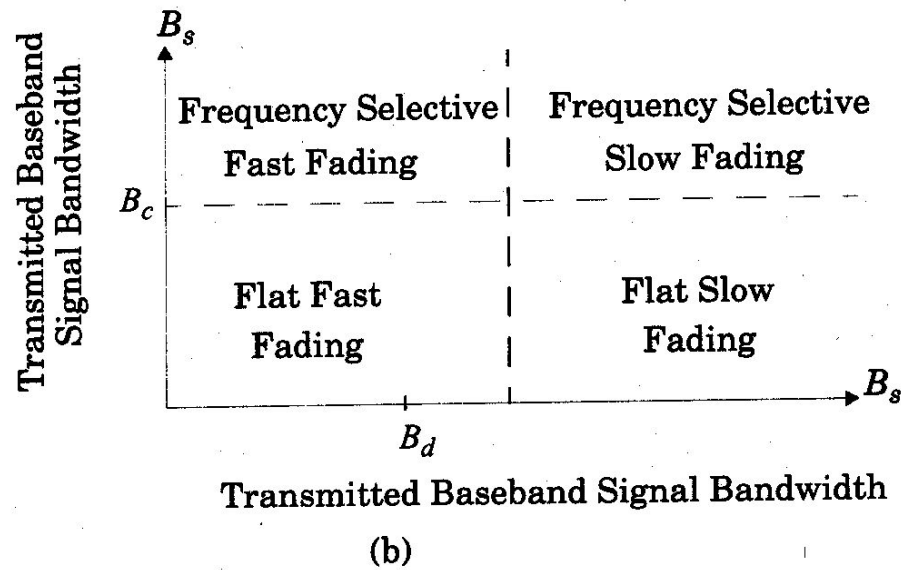
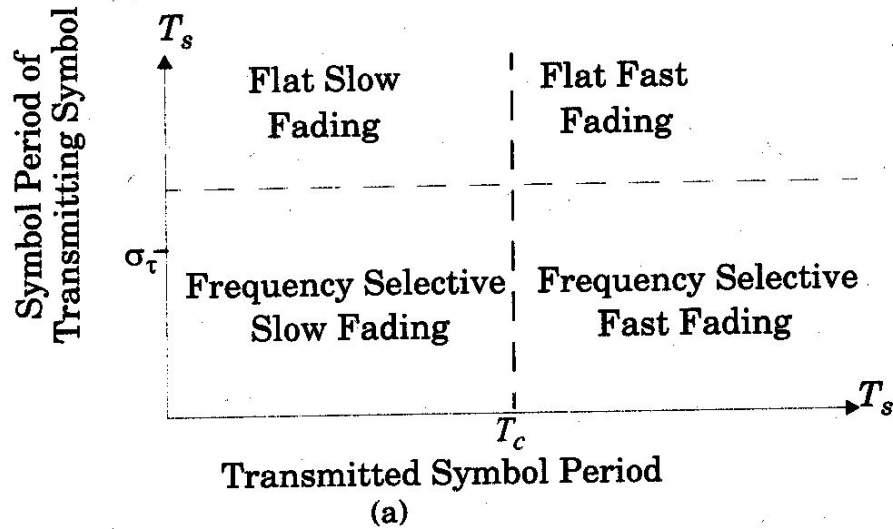
$$B_s < B_d$$

- Slow Fading: The channel impulse response changes at a rate much slower than the transmitted baseband signal $s(t)$.
 - The Doppler spread of the channel is much less than the bandwidth of the baseband signal.
- A signal undergoes slow fading if

$$T_s \ll T_c$$

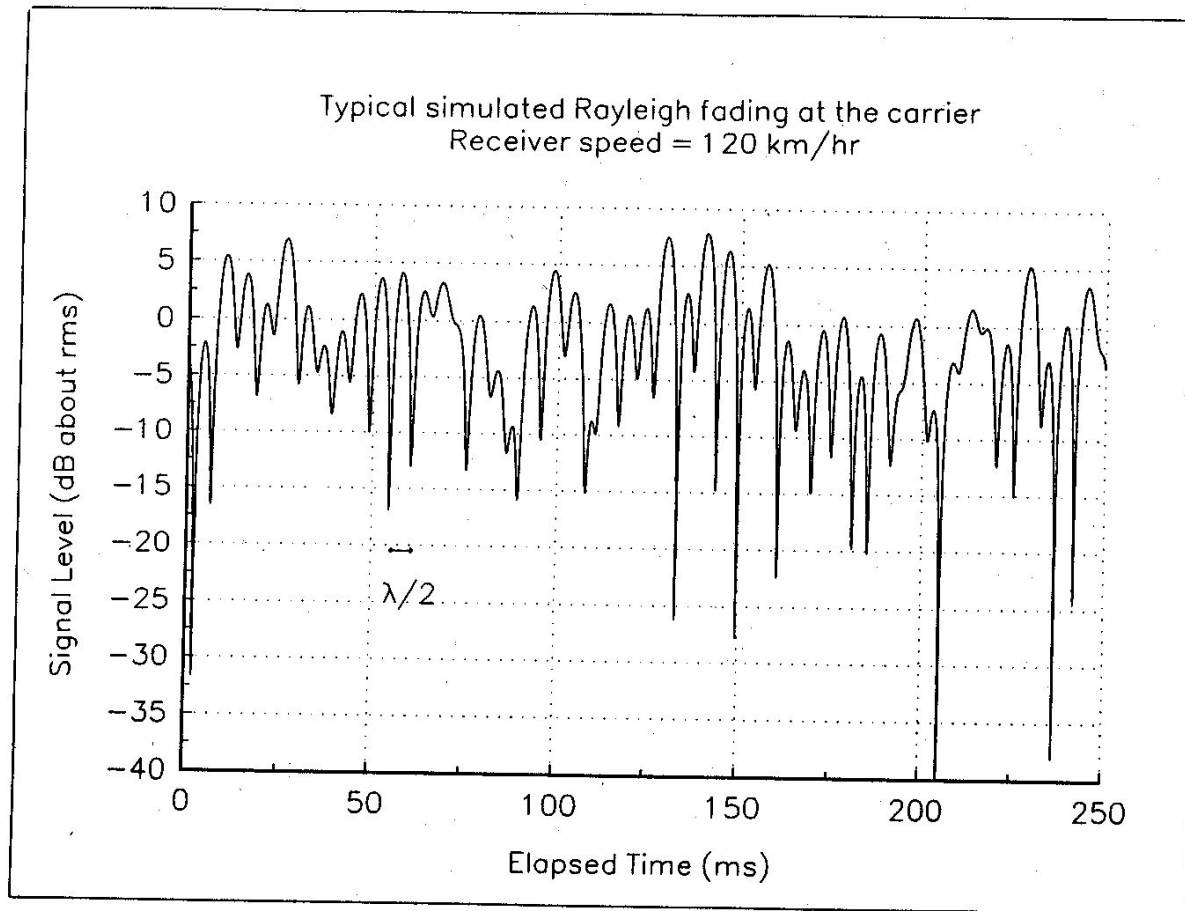
and

$$B_s \gg B_D$$



4.6 Rayleigh and Ricean Distributions

- Rayleigh Fading Distribution
 - The sum of two quadrature Gaussian noise signals



- Consider a carrier signal at frequency ω_0 and with an amplitude a

$$s(t) = a \exp(j\omega_0 t)$$

- The received signal is the sum of n waves

$$s_r(t) = \sum_{i=1}^n a_i \exp(j\omega_0 t + \theta_i) = r \exp[j(\omega_0 t + \theta)] = r \exp(j\theta) \exp(j\omega_0 t)$$

where

$$r \exp(j\theta) = \sum_{i=1}^n a_i \exp(j\theta_i)$$

define

$$r \exp(j\theta) = \sum_{i=1}^n a_i \cos(\theta_i) + j \sum_{i=1}^n a_i \sin(\theta_i) = x + jy$$

We have

$$x \equiv \sum_{i=1}^n a_i \cos(\theta_i) = r \cos(\theta) \quad \text{and} \quad y \equiv \sum_{i=1}^n a_i \sin(\theta_i) = r \sin(\theta)$$

- It can be assumed that x and y are Gaussian random variables with mean equal to zero due to the following reasons
 - n is usually very large.
 - The individual amplitude a_i are random.
 - The phases θ_i have a uniform distribution.
- Because x and y are independent random variables, the joint distribution $p(x,y)$ is

$$p(x, y) = p(x)p(y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right)$$

- The distribution $p(r, \theta)$ can be written as a function of $p(x, y)$

$$p(r, \theta) = |J| p(x, y)$$

$$|J| = \begin{vmatrix} \partial x / \partial r & \partial x / \partial \theta \\ \partial y / \partial r & \partial y / \partial \theta \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r$$

- We have

$$p(r, \theta) = \frac{r}{2\pi\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right)$$

- The Rayleigh distribution has a pdf given by

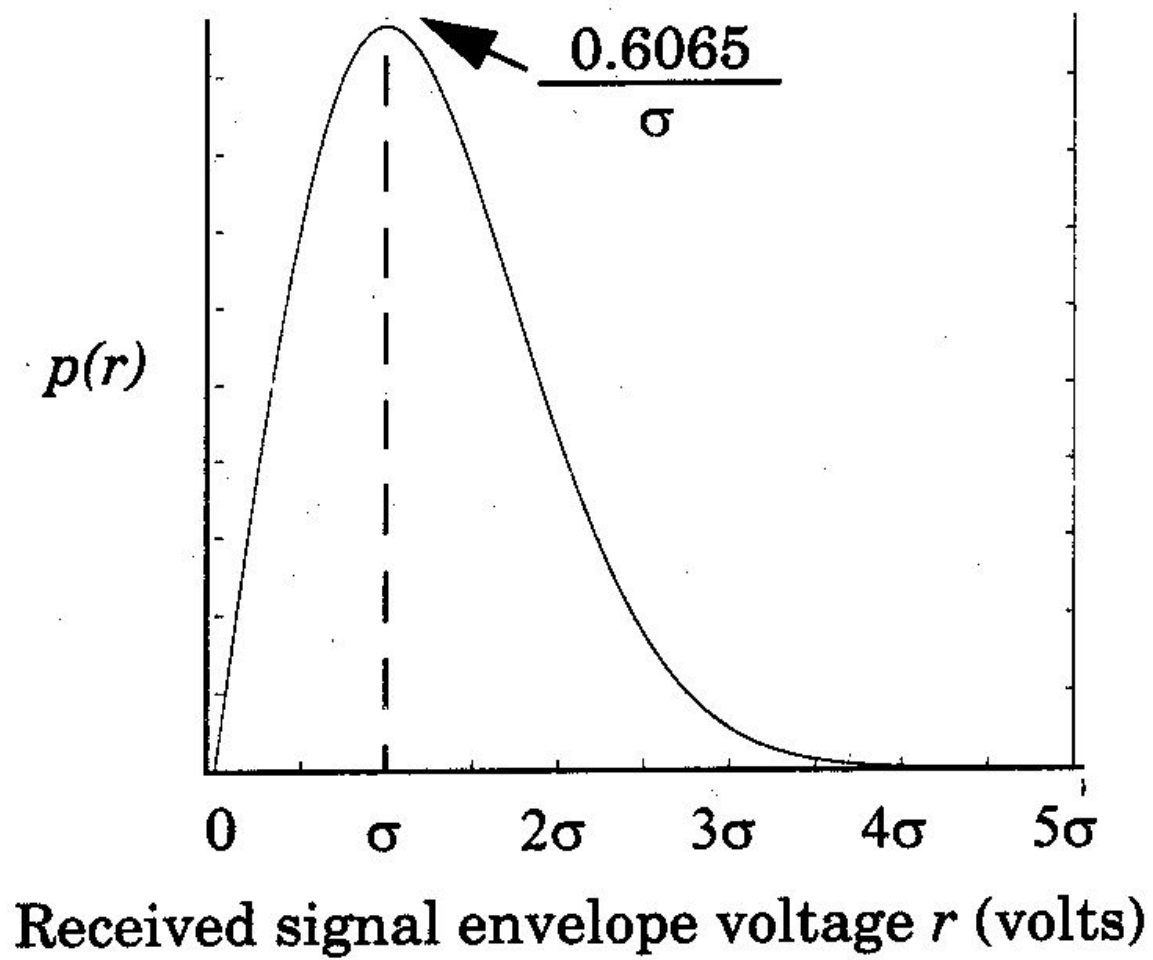
$$p(r) = \int_0^{2\pi} p(r, \theta) d\theta = \begin{cases} \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) & r \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- pdf of Rayleigh distribution

$$p(r) = \begin{cases} \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) & 0 \leq r \leq \infty \\ 0 & r < 0 \end{cases}$$

σ : rms value of the received signal before envelop detection

σ^2 : time - average power of the received signal before envelop detection



- Cumulative distribution function (CDF)

$$P(R) = \Pr(r \leq R) = \int_0^R p(r)dr = 1 - \exp\left(-\frac{R^2}{2\sigma^2}\right)$$

- The mean value of the Rayleigh distribution is given by

$$r_{mean} = E[r] = \int_0^\infty rp(r)dr = \sigma \sqrt{\frac{\pi}{2}} = 1.2533\sigma$$

- The variance of the Rayleigh distribution is given by

$$\begin{aligned}\sigma_r^2 &= E[r^2] - E^2[r] = \int_0^\infty r^2 p(r)dr - \frac{\sigma^2 \pi}{2} \\ &= \sigma^2 \left(2 - \frac{\pi}{2}\right) = 0.4292\sigma^2\end{aligned}$$

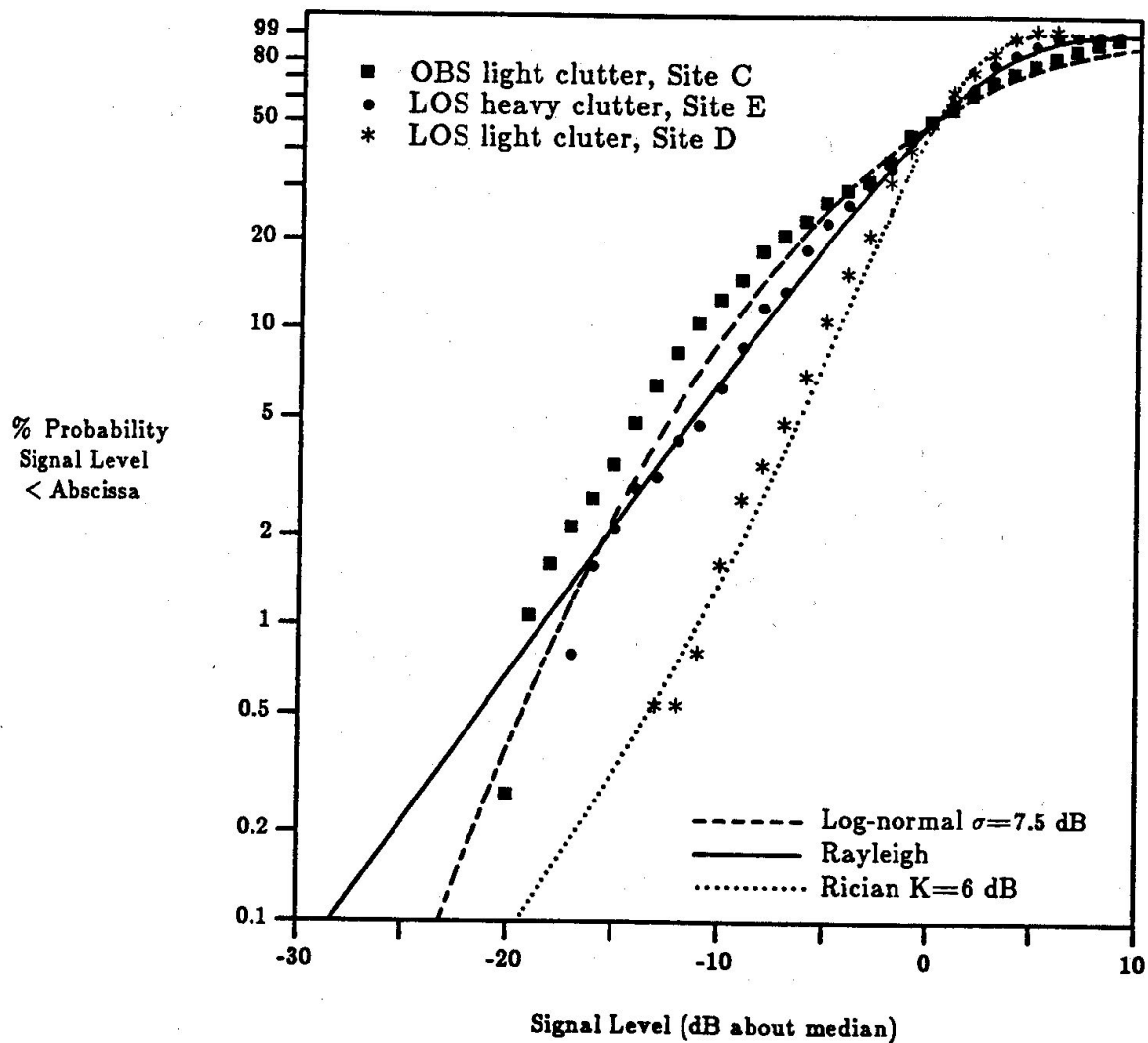


Figure 4.17

Cumulative distribution for three small-scale fading measurements and their fit to Rayleigh, Rician, and log-normal distributions [From [Rap89] © IEEE].

- **Ricean Fading Distribution:** When there is a dominant stationary (non-fading) signal component present, such as a line-of-sight propagation path, the small-scale fading envelope distribution is Ricean.

Scattered waves

Direct wave

$$s_r(t) = r \exp[j(\omega_0 t + \theta)] + A \exp(j\omega_0 t)$$

$$= [(x + A) + jy] \exp(j\omega_0 t)$$

$$r^2 = (x + A)^2 + y^2$$

$$x + A = r \cos \theta$$

$$y = r \sin \theta$$

- By following similar steps described in Rayleigh distribution, we obtain

$$p(r) = \begin{cases} \frac{r}{\sigma^2} \exp\left(-\frac{r^2 + A^2}{2\sigma^2}\right) I_0\left(\frac{Ar}{\sigma^2}\right) & A \geq 0, r \geq 0 \\ 0 & r < 0 \end{cases}$$

where

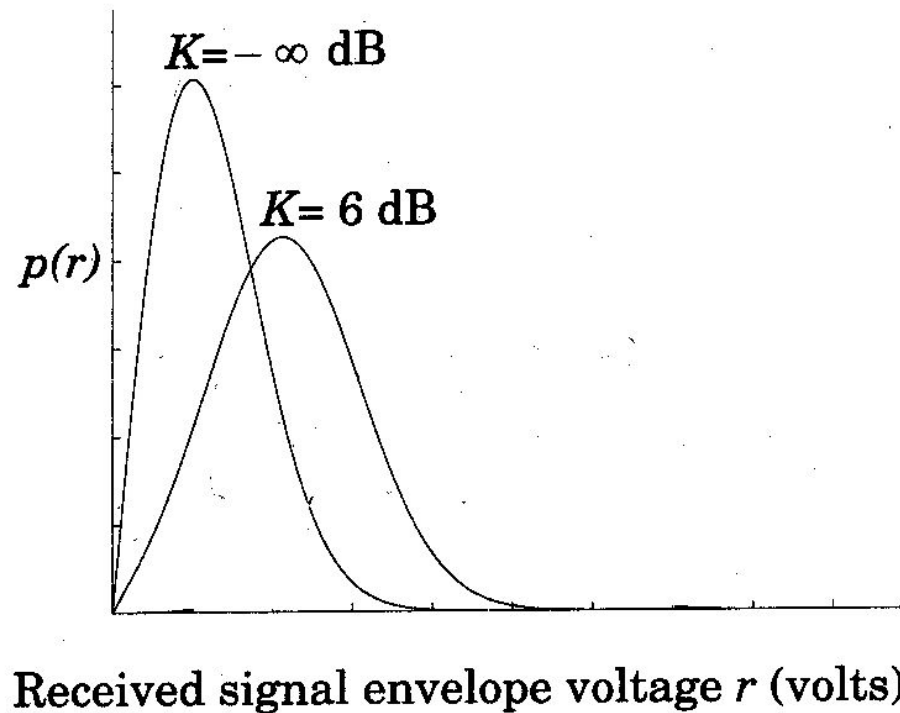
$$I_0\left(\frac{Ar}{\sigma^2}\right) = \frac{1}{2\pi} \int_0^{2\pi} \exp\left(\frac{Ar \cos \theta}{\sigma^2}\right) d\theta$$

is the modified Bessel function of the first kind and zero-order.

- The Ricean distribution is often described in terms of a parameter K which is defined as the ratio between the deterministic signal power and the variance of the multipath. It is given by $K = A^2 / (2\sigma^2)$ or in terms of dB

$$K(\text{dB}) = 10 \log \frac{A^2}{2\sigma^2} \quad \text{dB}$$

- The parameter K is known as the Ricean factor and completely specifies the Ricean distribution.
- As $A \rightarrow 0$, we have $K \rightarrow -\infty$ dB. The dominant path decrease in amplitude, the Ricean distribution degenerates to a Rayleigh distribution.



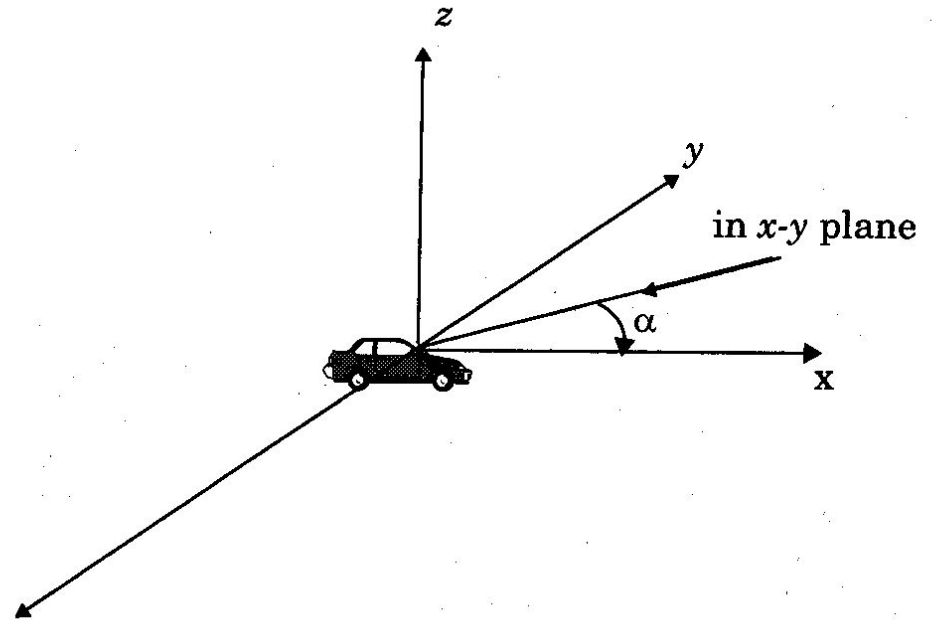
Statistical Models for Multipath Fading Channels

4.7.1 Clarke's Models for Flat Fading

- Clark developed a model where the statistical characteristics of the electromagnetic fields of the received signal are deduced from scattering.
- The model assumes a fixed transmitter with a vertically polarized antenna.
- The received antenna is assumed to comprise of N azimuthal plane waves with arbitrary carrier phase, arbitrary angle of arrival, and each wave having equal average amplitude.
- Equal amplitude assumption is based on the fact that in the absence of a direct line-of-sight path, the scattered components arriving at a receiver will experience similar attenuation over small-scale distance.

- Doppler shift due to the motion of the receiver.
- Assume no excess delay due to multipath.
 - Flat fading assumption.
- For the n th wave arriving at an angle α_n to the x -axis, the Doppler shift is given by

$$f_n = \frac{v}{\lambda} \cos \alpha_n$$



- The vertically polarized plane waves arriving at the mobile have E field components given by (assume a single tone is transmitted)

$$E_z(t) = E_0 \sum_{n=1}^N C_n \cos(2\pi f_c t + \theta_n)$$

E_0 : real amplitude of local average E - field (constant)

C_n : real random variable representing the amplitude of n th arriving wave.

f_c : carrier frequency.

θ_n : random phase of the n th arriving wave.

- The random arriving phase is given by

- The amplitude of E-field is normalized such that

$$\sum_{i=1}^N \overline{C_n^2} = 1$$

- $E_z(t)$ can be modeled as a Gaussian random process if N is sufficient large.
- Since the Doppler shift is very small when compared to the carrier frequency, the three field components may be modeled as narrow band random process.

$$E_z(t) = T_c(t) \cos(2\pi f_c t) + T_s(t) \sin(2\pi f_c t)$$

where

$$T_c(t) = E_0 \sum_{n=1}^N C_n \cos(2\pi f_n t + \phi_n)$$

$$T_s(t) = E_0 \sum_{n=1}^N C_n \sin(2\pi f_n t + \phi_n)$$

- $T_c(t)$ and $T_s(t)$ are Gaussian random processes which are denoted as T_c and T_s , respectively.

- $T_c(t)$ and $T_s(t)$ are uncorrelated zero-mean Gaussian random variable with equal variance given by

$$\overline{T_c^2} = \overline{T_s^2} = \overline{|E_z|^2} = E_0^2 / 2$$

- The envelope of the received E-field is given by

$$|E_z(t)| = \sqrt{T_c^2(t) + T_s^2(t)} = r(t)$$

- It can be shown that the random received signal envelope r has a Rayleigh distribution given by

$$p(r) = \begin{cases} \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) & 0 \leq r \leq \infty \\ 0 & r < 0 \end{cases}$$

where $\sigma^2 = E_0^2 / 2$

- Let $p(\alpha)d\alpha$ denote the function of the total incoming power within $d\alpha$ of the angle α , and let A denote the average received power with respect to an isotropic antenna.
- As $N \rightarrow \infty$, $p(\alpha)d\alpha$ approached a continuous distribution.
- If $G(\alpha)$ is the azimuthal gain pattern of the mobile antenna as a function of the angle of arrival, the total received power can be expressed as

$$P_r = \int_0^{2\pi} AG(\alpha)p(\alpha)d\alpha$$

- The instantaneous frequency of the received signal arriving at an angle α is given by:

$$f(\alpha) = f = \frac{v}{\lambda} \cos(\alpha) + f_c = f_m \cos \alpha + f_c$$

where f_m is the maximum Doppler shift.

- If $S(f)$ is the power spectrum of the received signal, the differential variation of received power with frequency is given by

$$S(f) |df| = A[p(\alpha)G(\alpha) + p(-\alpha)G(-\alpha)]d\alpha|$$

- Differentiation $f = f_m \cos \alpha + f_c$

$$\frac{|df|}{|d\alpha|} = f_m |\sin \alpha| \quad \Rightarrow \quad |df| = |d\alpha| \sin \alpha f_m$$

On the other hand, we have $\alpha = \cos^{-1} \left[\frac{f - f_c}{f_m} \right]$

- This implies

$$\sin \alpha = \sqrt{1 - \left(\frac{f - f_c}{f_m} \right)^2}$$

- Finally, we have

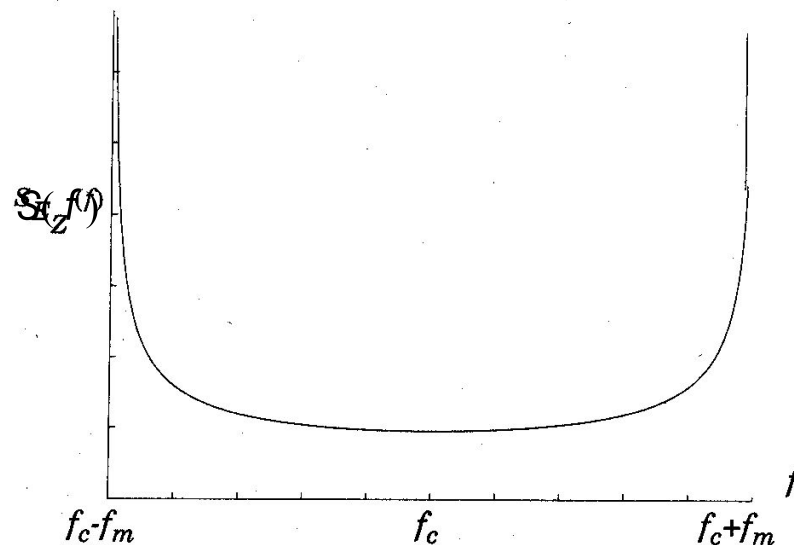
$$S(f) = \frac{A[p(\alpha)G(\alpha) + p(-\alpha)G(-\alpha)]}{f_m \sqrt{1 - \left(\frac{f - f_c}{f_m}\right)^2}}$$

where $S(f) = 0$, $|f - f_c| > f_m$

- The spectrum is centered on the carrier frequency and is zero outside the limits $f_c \pm f_m$.
- Each of the arriving waves has its own carrier frequency (due to its direction of arrival) which is slightly offset from the center frequency.

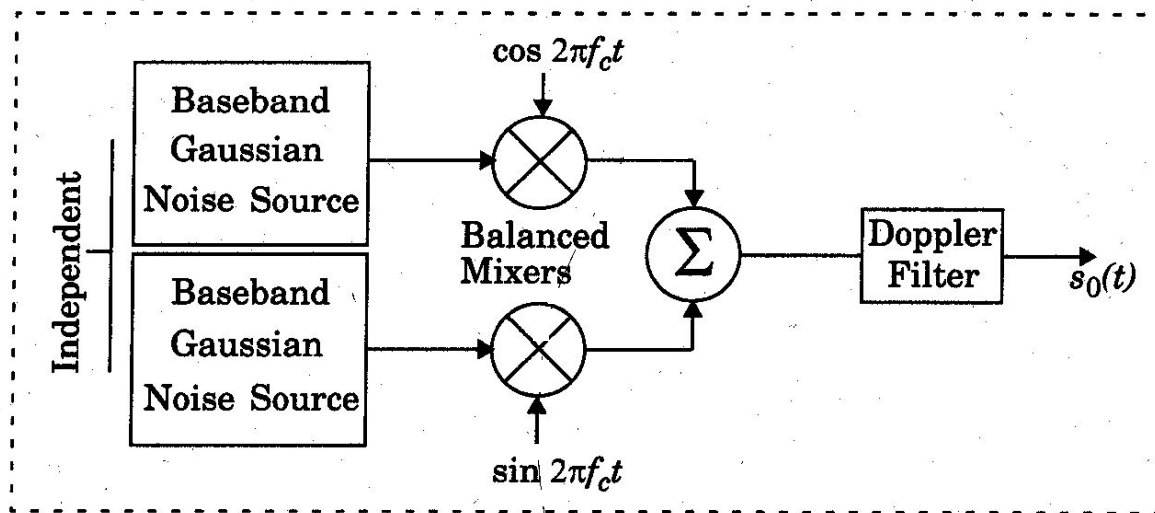
- Vertical $\lambda / 4$ antenna ($G(\alpha) = 1.5$).
- Uniform distribution $p(\alpha) = 1/(2\pi)$ over 0 to 2π .
- The output spectrum

$$S(f) = \frac{1.5}{\pi f_m \sqrt{1 - \left(\frac{f - f_c}{f_m} \right)^2}}$$

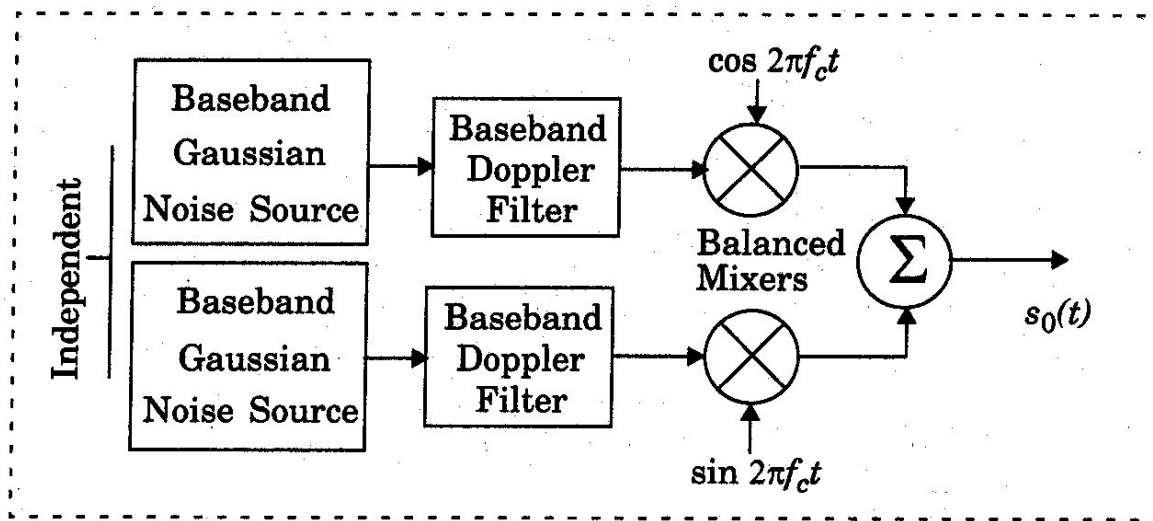


4.7.2 Simulation of Clarke Fading Model

- Produce a simulated signal with spectral and temporal characteristics very close to measured data.
- Two independent Gaussian low pass noise are used to produce the in-phase and quadrature fading branches.
- Use a spectral filter to sharp the random signal in the frequency domain by using fast Fourier transform (FFT).
- Time domain waveforms of Doppler fading can be obtained by using an inverse fast Fourier transform (IFFT).

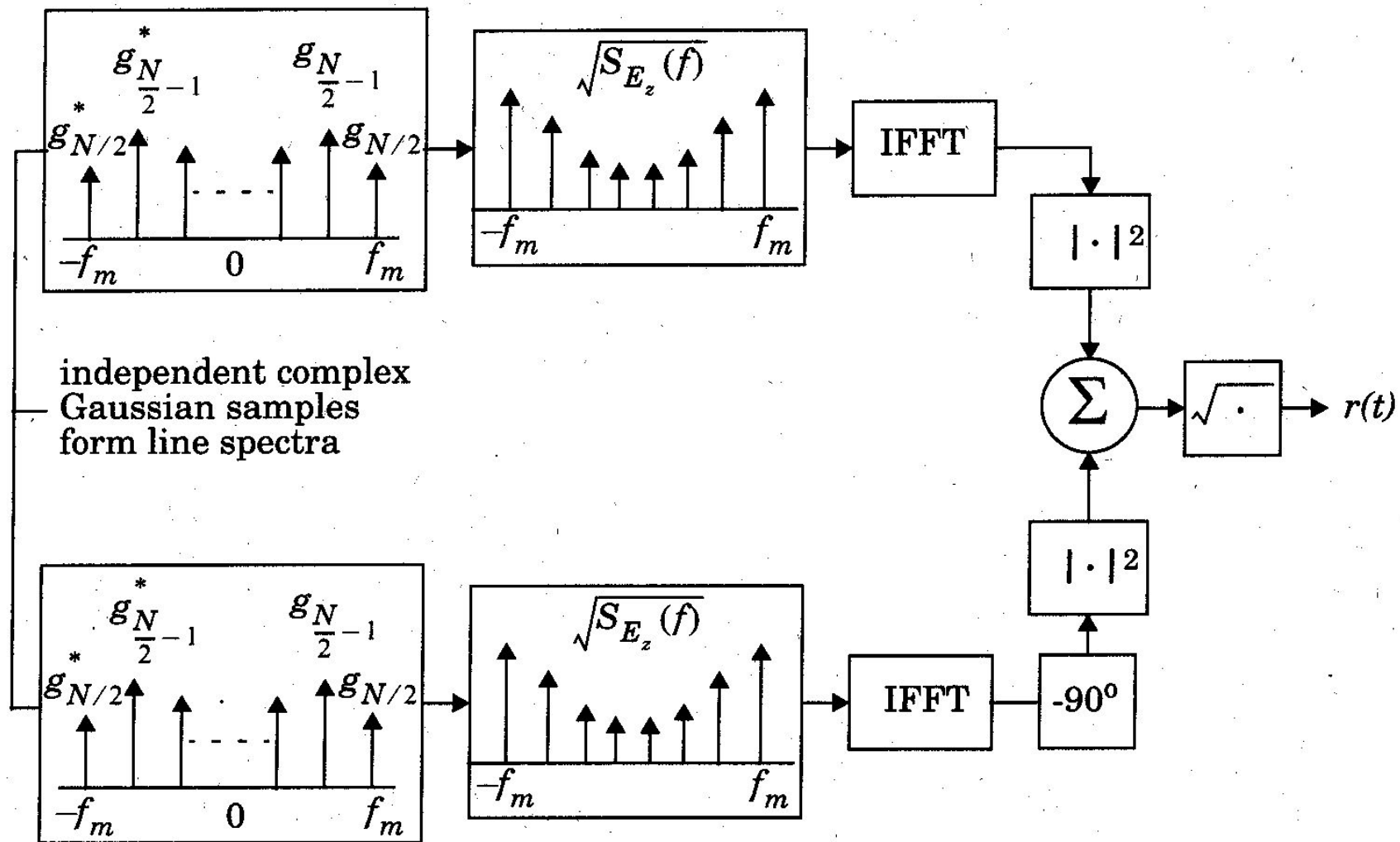


(a)

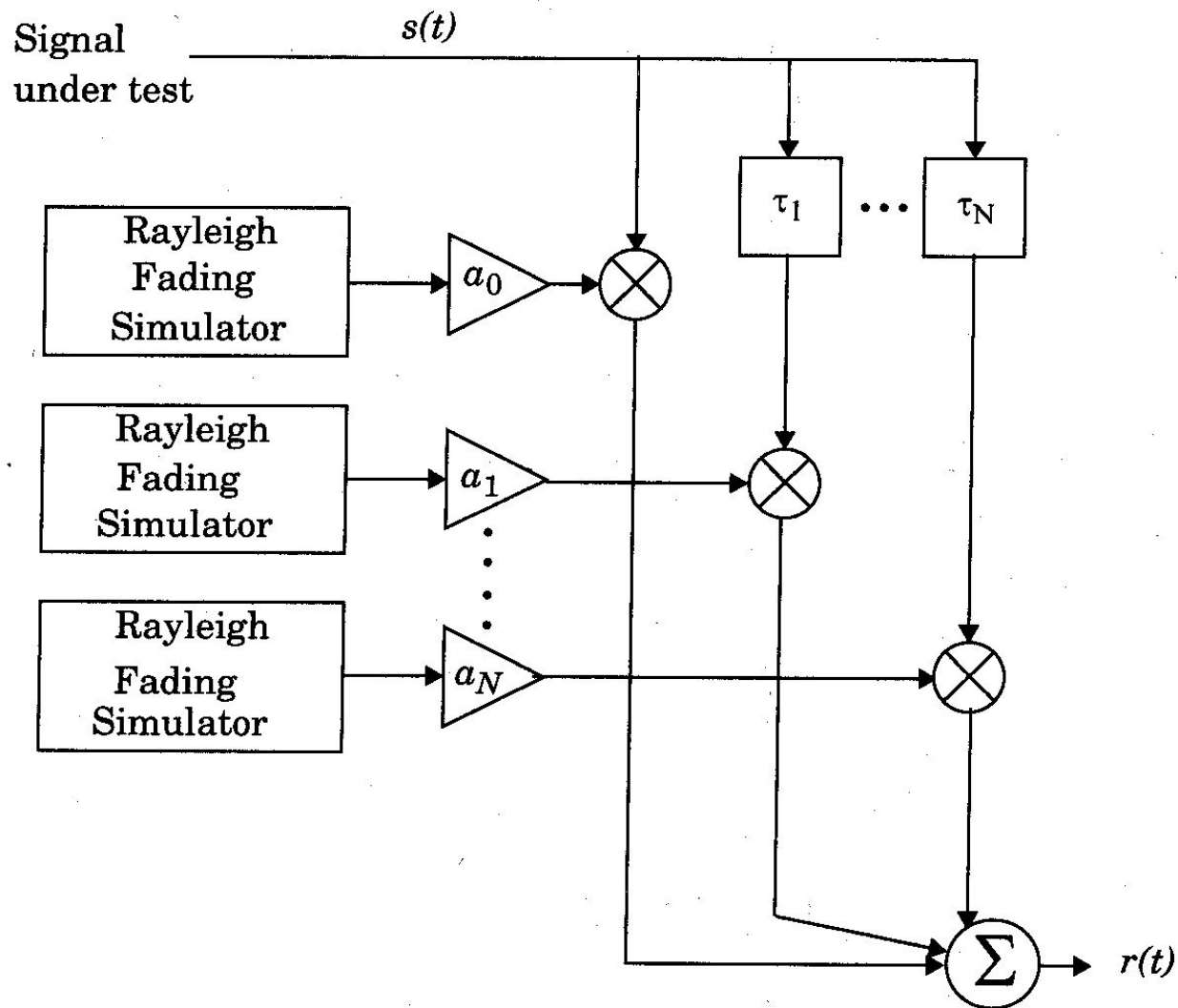


(b)

- Smith simulator using N carriers to generate fading signal
 1. Specify the number of frequency domain points N used to represent $\sqrt{S(f)}$ and the maximum Doppler frequency shift f_m .
 2. Compute the frequency spacing between adjacent spectral lines as $\Delta f = 2f_m / (N - 1)$. This defines the time duration of a fading waveform, $T = 1 / \Delta f$.
 3. Generate complex Gaussian random variables for each of the $N/2$ positive frequency components of the noise source.
 4. Construct the negative frequency components of the noise source by conjugating positive frequency and assigning these at negative frequency values.
 5. Multiply the in-phase and quadrature noise sources by the fading spectrum $\sqrt{S(f)}$.
 6. Perform an IFFT on the resulting frequency domain signal from the in-phase and quadrature arms, and compute the sum of the squares of each signal.
 7. Take the square root of the sum.



- Frequency selection fading model



4.7.3 Level Crossing and Fading Statistics

- The level crossing rate (LCR) is defined as the expected rate at which the Rayleigh fading envelope crosses a specified level in a positive-going direction.
- Useful for designing error control codes and diversity.
- Relate the time rate of change of the received signal to the signal level and velocity of the mobile.
- The number of level crossing per second to the level R is given by

$$N_R = \int_0^\infty \dot{r} p(R, \dot{r}) d\dot{r} = \sqrt{2\pi} f_m \rho e^{-\rho^2} \quad (\text{A})$$

$\dot{r}$: time derivation of $r(t)$ (slope)

$p(R, \dot{r})$: joint density function of r and $\dot{r}$ at $r = R$.

f_m : maximum Doppler frequency

$\rho = R / R_{rms}$: value of the specified level R , normalized to the rms amplitude of the fading envelope.

- Average fading duration is defined as the average period of time for which the received signal is below a specified level R .
- For a Rayleigh Fading signal, this is given by

$$\bar{\tau} = \frac{1}{N_R} \Pr[r < R] \quad (\text{B})$$

with

$$\Pr[r < R] = \frac{1}{T} \sum_i \tau_i$$

where τ_i is the duration of the fade and T is the observation interval.

- For Rayleigh distribution

$$\Pr[r < R] = \int_0^R p(r) dr = 1 - \exp(-\rho^2) \quad (\text{C})$$

- Average fading duration, (using (A), (B), (C))

$$\bar{\tau} = \frac{e^{\rho^2} - 1}{\rho f_m \sqrt{2\pi}}$$

- The average duration of a signal fading helps determine the most likely number of signaling bits that may be lost during a fade.
- Average fade duration primarily depends upon the speed of the mobile, and decreases as the maximum Doppler frequency f_m becomes large.